Do **not** write solutions on this page.

11. [Maximum mark: 17]

(a) Find the first four terms in the binomial expansion of $\sqrt{1+5x}$ in ascending powers of x. [4]

Consider the expression $(1+px)(1+qx)^{-1}$, where $p, q \in \mathbb{Q}$.

(b) Find the expansion of $(1+px)(1+qx)^{-1}$ in ascending powers of x, up to and including the term in x^2. [3]

The expansions found in parts (a) and (b) are identical up to the first three terms, for a value of p and a value of q.

(c) Show that $q = \frac{5}{4}$. [4]

(d) The expression $\frac{1+px}{1+qx}$, with $p = \frac{15}{4}$ and $q = \frac{5}{4}$, can be used as an approximation for $\sqrt{1+5x}$ where $|x| < \frac{1}{5}$.

(i) Hence, by finding a suitable value for x, find the approximation for $\sqrt{1.2}$ in the form $\frac{m}{n}$, where $m, n \in \mathbb{Z}$.

(ii) Now consider the approximation for $\frac{\sqrt{5}}{2}$. Explain why the approximation for $\frac{\sqrt{5}}{2}$ is not as accurate as the approximation for $\sqrt{1.2}$. [6]